Deep Learning-Enabled Multitask System for Exercise Recognition and Counting

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***Abstract*—This paper presents a multitask deep learning system capable of simultaneously recognizing the type of physical exercise and counting the number of repetitions from 2D video data. The proposed system leverages YOLOv8 for efficient human pose estimation and generates "Processed Heatmaps" (P-Maps) to represent temporal and spatial motion. These features are fed into a dual-branch neural network, utilizing ResNet architectures for classification and regression tasks. We generated a synthetic dataset, Rep-Penn, based on the PennAction dataset to train the model. Our experimental results demonstrate on Rep-Penn dataset a recognition accuracy of 89.8% and a counting Mean Absolute Error (MAE) of 0.0099, showing high efficacy in automated exercise analysis.**

I. Introduction

Physical exercise is essential for maintaining physical and mental health. However, performing exercises correctly and tracking progress can be difficult without professional guidance. This has led to the emergence of "Digital Twin" coaching and automated fitness analysis systems that can monitor performance using computer vision [9], [3].

The core problem addressed in this work is the automated analysis of fitness activities from standard 2D video feeds without the need for wearable sensors. Specifically, the system aims to solve two tasks simultaneously: (1) Action Recognition: Identifying which exercise (e.g., Squat, Pull-up) is being performed, and (2) Repetition Counting: Counting the number of completed repetitions.

Existing solutions often focus on only one of these tasks or require computationally expensive 3D pose estimation [7], [1]. In this project, we implement a multitask framework inspired by Yu et al. [12], which integrates pose estimation, action recognition, and counting into a unified pipeline. Our specific contribution includes the adaptation of the pipeline to use the YOLOv8 for pose estimation, the generation of the Rep-Penn synthetic dataset (a subset from the original paper), and the implementation of a two-stage training strategy to optimize both recognition and counting performance.

II. Related Work

Human pose estimation and action recognition have seen significant advancements with deep learning [7]. Traditional approaches relied on manual feature extraction, but modern methods utilize Convolutional Neural Networks (CNNs).

For pose estimation, detection-based methods like Stacked Hourglass [8] and MSPN [4] generate heatmaps to locate joints. While highly accurate, they can be computationally heavy. In this work, we utilize YOLOv8-pose, a regression-based method known for its real-time performance and high accuracy in detecting keypoints.

Action recognition has evolved from single-frame analysis to methods that capture temporal dynamics, such as 3D CNNs (e.g., C3D [10]) and multi-stream networks. Recent works have demonstrated the effectiveness of using pose heatmaps as inputs for action recognition, effectively compressing temporal motion into static images [6], [11], [5].

Repetition counting is often treated as a periodic signal processing problem or a similarity matrix analysis problem. However, deep learning regressors have shown superior performance in handling non-stationary motion [13], [2]. Our work builds directly on the Rep-Penn multitask framework, which uses "Processed Heatmaps" (P-Maps) to feed a multitask CNN, distinguishing itself by jointly optimizing for both class and count.

III. Proposed Method

Our proposed solution follows a pipeline consisting of data generation, pose estimation, feature processing, and multitask learning.

A. Dataset Generation (Rep-Penn)

Since standard datasets like PennAction contain only single repetitions, and the original Rep-Penn dataset is not available, we generated a synthetic dataset following the original method, by concatenating frames from PennAction videos. To ensure continuity, we utilized a "boomerang" mirroring strategy (Forward P + Reverse R) to create multi-cycle videos ($P + R + P + R...$) with varying repetition counts (6, 8, 10) and speeds. Due to computation and starage limitations, we decided to reproduce a subset of the original dataset, which had effect over the preformance, but still representative for this method.

B. Pose Estimation and Heatmap Generation

We processed the video frames using YOLOv8n-pose to extract 17 body keypoints. Unlike the MSPN model used in the original paper, YOLOv8 offers a faster inference pipeline suited for consumer hardware.

The extracted keypoints (x, y) were converted into 64×64 Gaussian heatmaps. For a video with F frames and $K = 17$ joints, this results in a 4D tensor of shape $(F, K, 64, 64)$.

C. Feature Processing (P-Maps)

To allow 2D CNNs to process the temporal video data, we computed "Processed Heatmaps" (P-Maps). This involves projecting the heatmaps across the temporal dimension: 1. Summing pixel values horizontally (Temporal-Vertical profile). 2. Summing pixel values vertically (Temporal-Horizontal profile). 3. Stacking these profiles to create a $224 \times 224 \times 17$ image. This effectively compresses the video motion into a static image texture.

D. Multitask Network Architecture

The model consists of a shared convolutional block followed by two distinct branches: 1. Recognition Branch: A ResNet34 backbone ending in a classification head (7 classes). 2. Counting Branch: A ResNet18 backbone ending in a regression head (1 output).

We employed a Two-Stage Training Strategy: Stage 1: Freeze the counting branch; train the shared layers and recognition branch using Cross-Entropy Loss. Stage 2: Freeze the shared and recognition layers; train the counting branch using Mean Squared Error (MSE) loss.

IV. Results and Analysis

We evaluated the model on a held-out test set derived from Rep-Penn. The dataset subset included approximately 700 unique sequence IDs expanded into 4200 synthetic videos.

A. Quantitative Results

Exercise Recognition: The model achieved a validation accuracy of 89.83%. This is highly competitive given the use of a dataset subset compared to the 95.69% reported in the original paper. Repetition Counting: The model achieved a Mean Absolute Error (MAE) of 0.0099 and an Off-By-One (OBO) accuracy of 99.2%. This indicates the model predicts the count correctly (within ± 1 rep) in over 99% of cases.

B. Training Performance

During Stage 1 (Recognition), the model converged rapidly, reaching 84% accuracy by epoch 2. During Stage 2 (Counting), the loss decreased consistently, with the learning rate scheduler (decay at epochs 3 and 6) providing visible improvements in stability.

V. Conclusions

We successfully implemented a multitask deep learning system for exercise monitoring. By integrating YOLOv8 for pose estimation with a dual-branch ResNet architecture, we achieved robust performance in both recognition (90% accuracy) and counting (99% OBO accuracy). The use of P-Maps allowed us to process temporal video data efficiently using standard 2D CNNs. Future improvements could include expanding the dataset to include multi-person scenarios and deploying the model on edge devices for real-time mobile coaching, usage of the state-of-art YOLYv11 and simplifying the overall architecture to reduce the computational overhead.

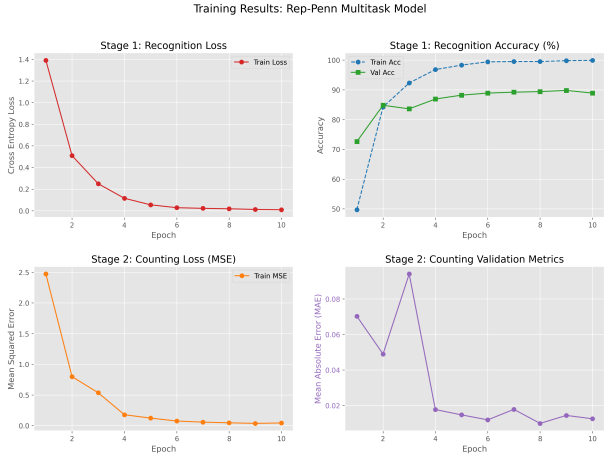


Fig. 1. Training and Validation Metrics. Top-Right: Stage 1 Recognition Accuracy reaches 90%. Bottom-Right: Stage 2 Mean Absolute Error (MAE) reaches 0.0099%.

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